

Module 09 – Social Engineering

Hands-On Lab Report

Candidate	Dhruva Khare
Module	09 – Social Engineering
Course	CEHv13 – Certified Ethical Hacker
Lab Duration	1 hour 47 minutes
Lab Environment	Parrot Security (attacker) / Windows 11 (victim) – isolated training network
Assessment Result	2 / 2 — Passed
Report Date	August 16, 2026

1. Objective

This report documents the practical exercises completed for Module 09 (Social Engineering) of the CEHv13 curriculum. The objective of the lab was to gain hands-on familiarity, within an authorized and isolated training environment, with two categories of social-engineering tradecraft covered by the course:

- Credential harvesting using the Social-Engineer Toolkit (SET) on Parrot Security, targeting a lab-provided practice site ("Moviescope") and a Windows 11 victim VM.
- Recognition and reporting of phishing/spoofed websites and phishing emails using Netcraft and manual analysis, and identification of the geolocation data associated with a suspicious host.

All activity was performed strictly inside the vendor-supplied CEH lab range against lab-owned targets. No production systems, real user accounts, or third-party infrastructure were targeted.

2. Tools & Resources Used

- Social-Engineer Toolkit (SET) v8.0.3 — Parrot Security OS
- SET Website Attack Vector → Credential Harvester / Site Cloner
- Target clone: <http://www.moviescope.com> (lab practice site)
- Microsoft Outlook (web) — phishing email drafting/inspection exercise
- Netcraft Extension — malicious/phishing URL reporting and browser warning behavior
- IP/host geolocation lookup tool (Leaflet / OpenStreetMap-based)
- ChatGPT — used to review wording of a legitimate password-reset notice for comparison against phishing indicators

3. Procedure

3.1 Launching the Social-Engineer Toolkit

SET was launched on the Parrot Security machine. The toolkit reported that a newer release (8.1.3) was available against the installed version (8.0.3). From the main Social-Engineering Attacks menu, the ten available attack categories were reviewed, including Spear-Phishing, Website Attack Vectors, Infectious Media Generator, and others.

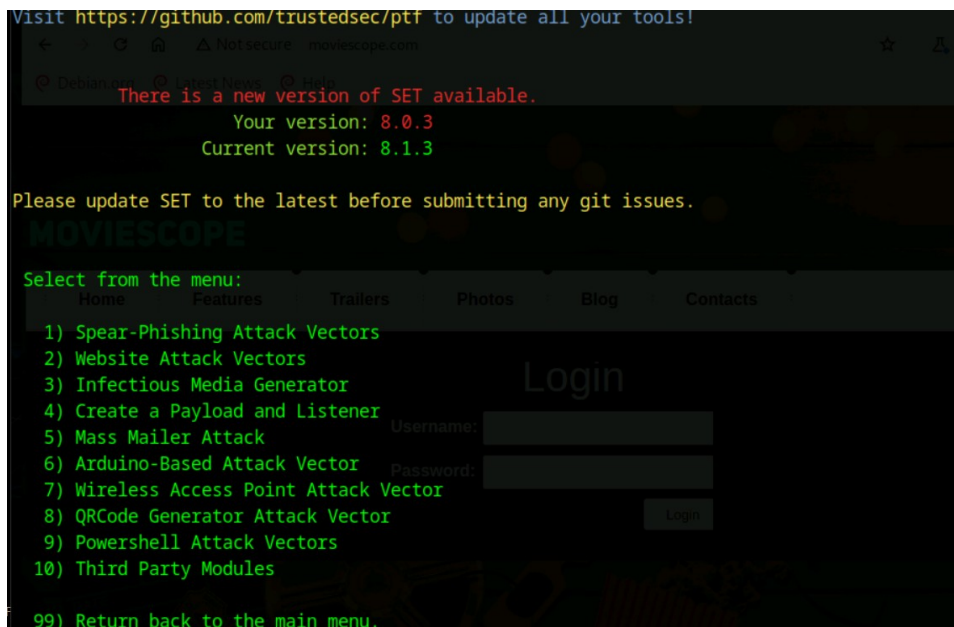


Figure 1 — SET main menu showing available social-engineering attack vectors.

3.2 Cloning the Target Site with the Credential Harvester

Within the Website Attack Vector module, the Credential Harvester / Site Cloner method was selected. SET was configured with the local listener IP address for the POST-back of harvested credentials, and the target URL (<http://www.moviescope.com>) was supplied for cloning. SET then cloned the site's login page so that any credentials entered by a victim on the fake page would be captured by the listener.

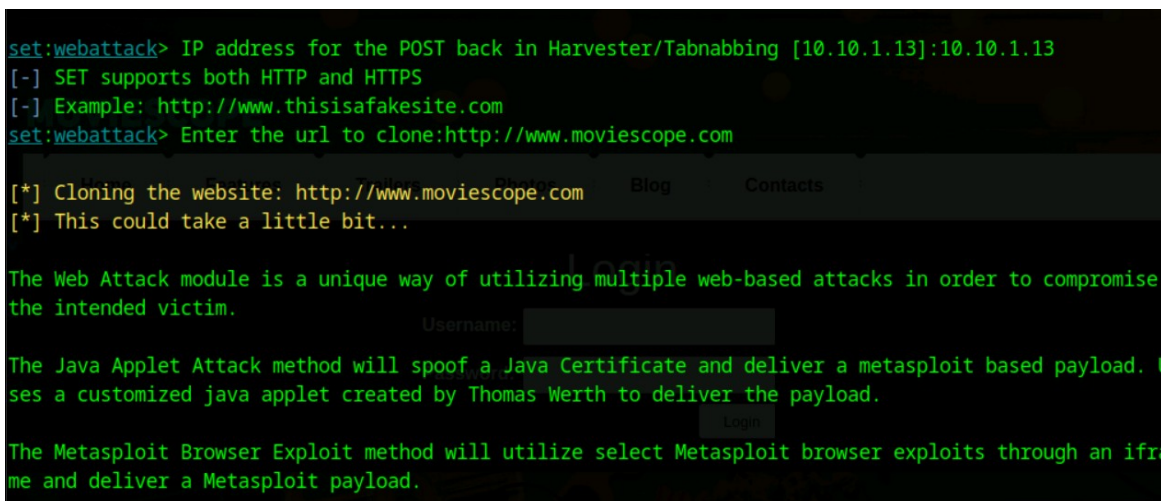


Figure 2 — SET cloning the Moviescope site and describing the Web Attack module options.

This exercise demonstrated how quickly a convincing fake login page can be produced and hosted, reinforcing why user awareness and URL verification are critical defenses against credential-harvesting attacks.

3.3 Phishing E-mail Analysis

A sample phishing e-mail impersonating the "Moviescope" service was drafted and reviewed in Outlook to study typical social-engineering pretexts. The message used urgency ("account temporarily suspended", "permanently disabled") and a call to action (click a link to "verify" account details) — both classic manipulation cues taught in the module.

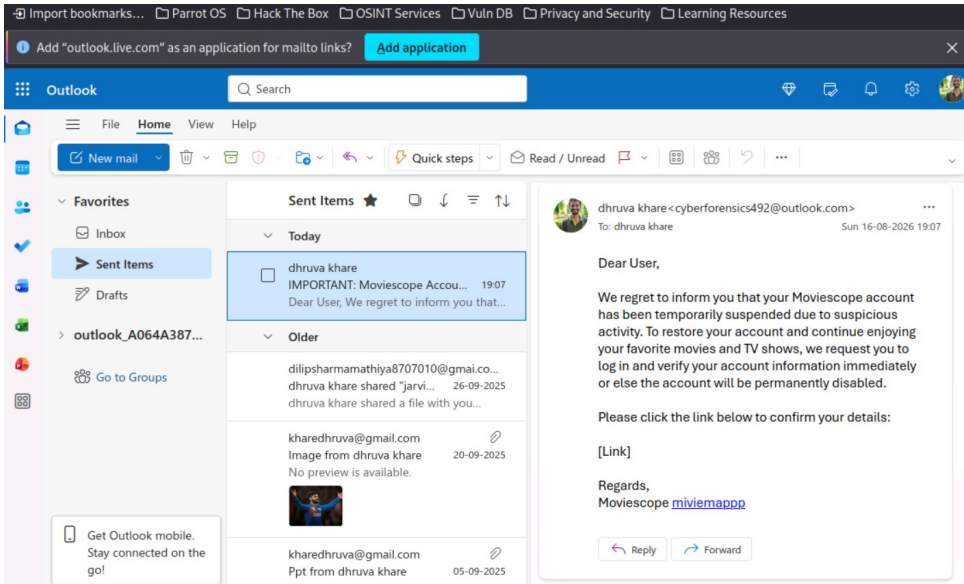


Figure 3 — Sample phishing e-mail (lab exercise) impersonating a streaming service to pressure the recipient into clicking a verification link.

3.4 Phishing/Malicious URL Detection with Netcraft

The Netcraft browser extension was used to inspect a test site's report. Netcraft's Site Report panel exposes metadata such as country, site rank, first-seen date, and hosting provider, and provides a built-in mechanism to report a URL as malicious/phishing directly from the toolbar.

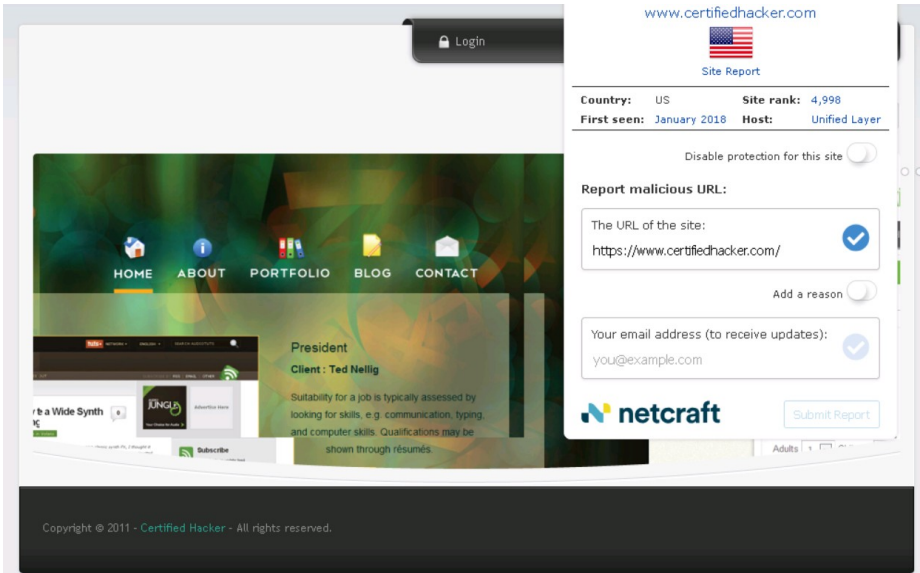


Figure 4 — Netcraft Site Report panel with the "Report malicious URL" workflow.

When Netcraft identifies a site as a phishing destination, it interposes a browser-level warning labeled "Suspected Phishing" before the page is allowed to load, blocking casual access to the reported page.

3.5 Host Geolocation Review

As part of validating the legitimacy of a host associated with the exercise, the advertised IP geolocation was compared against a multilaterated (measurement-based) location estimate. The advertised location (continental United States) and the multilaterated location did not align exactly, illustrating that IP geolocation data can be imprecise or spoofed and should not be relied upon as sole proof of a host's true origin.



Figure 5 — Comparison of advertised country vs. multilaterated location for the host under review.

3.6 Comparing Against a Legitimate Notice

For contrast, a legitimate-style Microsoft account security notice was reviewed. Genuine vendor communications typically avoid urgent threats, direct the recipient to navigate to the provider's site independently rather than clicking an embedded link, and explicitly warn against sharing passwords or verification codes — the opposite pattern from the phishing sample analyzed in section 3.3.

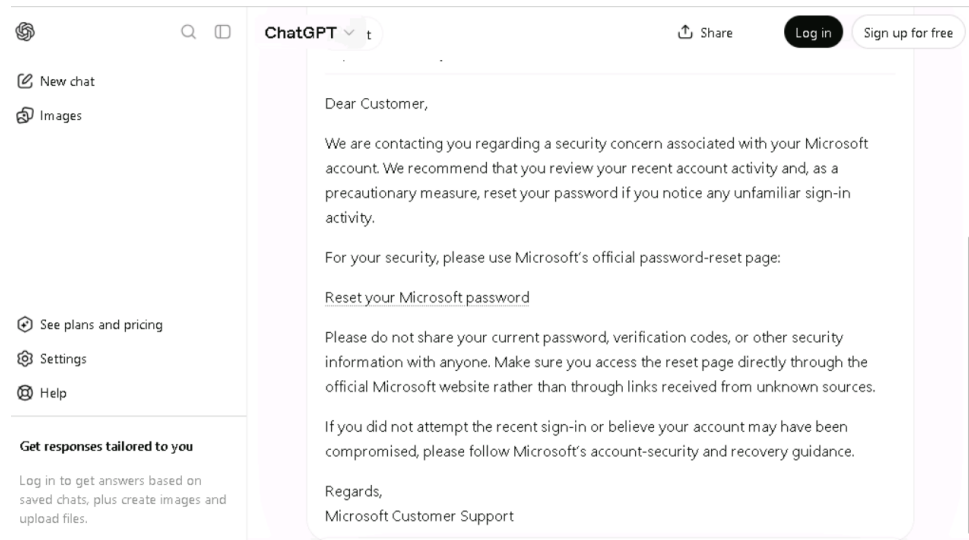


Figure 6 — Example of legitimate account-security guidance, used as a reference baseline against phishing indicators.

4. Assessment Results

The lab's knowledge-check questions were answered based on the exercises performed above:

Assessment Question	Answer / Result
Apart from Web Templates and Site Cloner, what is the third method that SET offers to deploy a credential-harvesting attack vector?	Custom Import
If Netcraft identifies any site as a phishing website, what message will Netcraft display on the user's web browser?	Suspected Phishing

Final score: 2 / 2 — Module 09 marked as Passed.

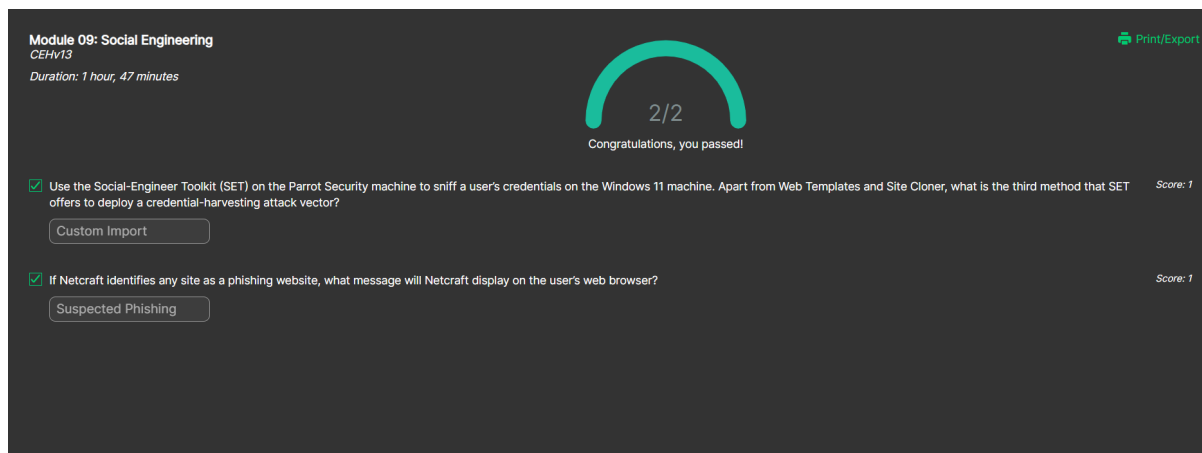


Figure 7 — Module 09 assessment summary confirming a 2/2 passing score.

5. Conclusion / Key Takeaways

- SET's Website Attack Vector can rapidly clone a legitimate site to harvest credentials, underscoring the need for users to verify URLs and use MFA rather than trusting page appearance alone.
- Phishing e-mails reliably rely on urgency and a pressured call-to-action; legitimate vendor notices instead direct users to navigate independently and never request sensitive data via a link.
- Netcraft's crowd-reported reputation data (Site Report + "Suspected Phishing" browser warning) is an effective control for flagging known-malicious pages before a user submits data.
- IP-based geolocation is only an approximation and can diverge from a host's true (multilaterated) location, so it should be one signal among several during an investigation, not definitive proof.

This exercise reinforced Module 09's core lesson: social engineering defeats technical controls by targeting human decision-making, so layered defenses — user training, URL/reputation checks, e-mail authentication, and MFA — remain essential complements to technical hardening.

— End of Report —